

A collection of historical artifacts is arranged on a light-colored surface. In the top left, a portion of a wooden chessboard with a checkered pattern and several chess pieces is visible. Below the chessboard, there are two medals: one with a red ribbon and a star-shaped emblem, and another with a blue ribbon and a star-shaped emblem. A pair of round-rimmed glasses with thin frames lies diagonally across the center. In the bottom left corner, a small, round, silver-colored compass is partially visible.

Multi-agent Coordination

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Outlines

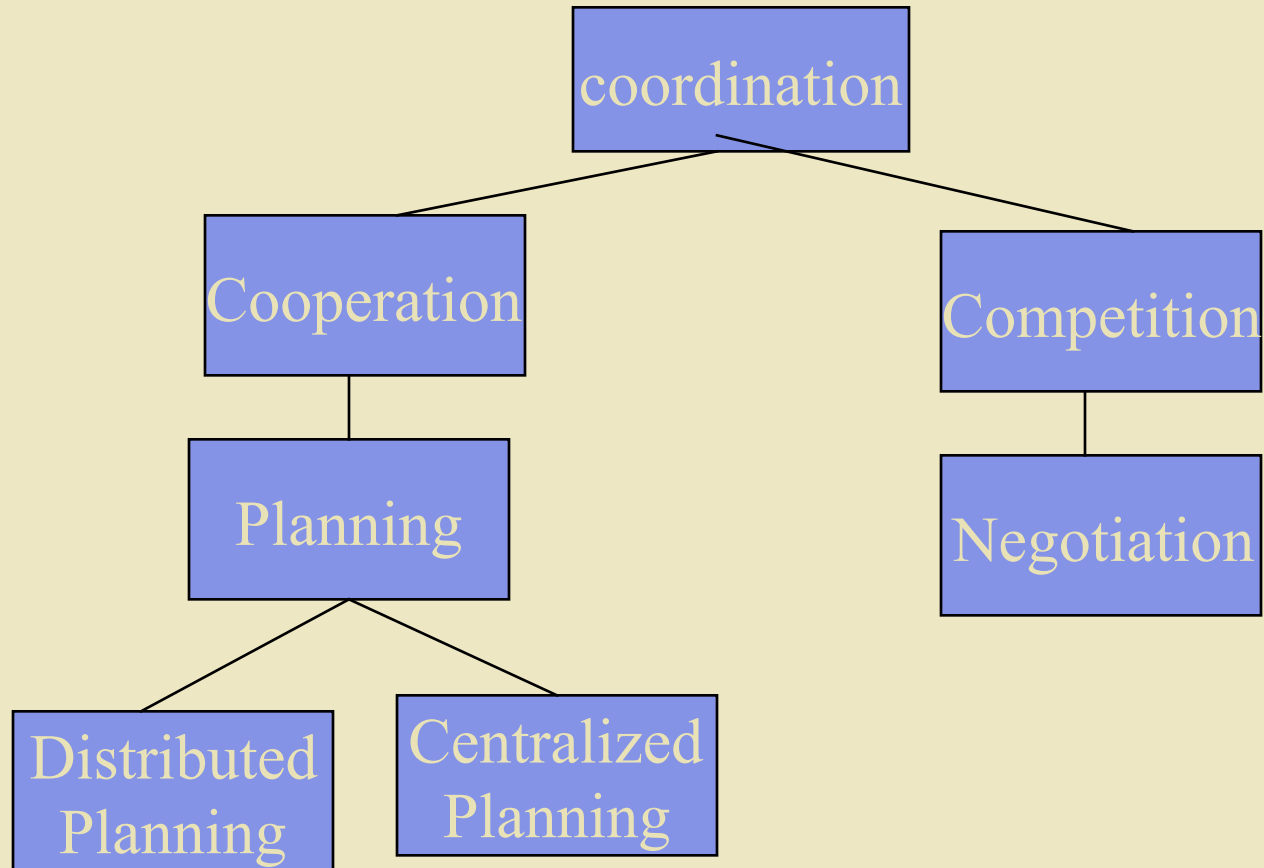
- Introduction
- ◆ Contract Net Protocol
- ◆ Task oriented domain negotiation mechanisms
- ◆ Trusted third party mediated game theoretic negotiation
- ◆ Market oriented resource allocations



What is coordination?

- ◆ Coordination is a coherent task assignment and execution.
- ◆ Coordination = Planning + Control + Communication
- ◆ Coordination = conflict resolving + resource sharing + efficiency enhancing
= Avoid deadlock+reduce resource contention+ avoid livelock

A taxonomy of coordination





Why is multi-agent coordination important?

- ◆ human complex problem solving are multi-agent in nature
 - distributed problem solving and
 - distributed artificial intelligence
- ◆ resolve conflicts among multi-agents
- ◆ prevent an anarchy or chaos
- ◆ satisfy global constraints and by working as a team, enhance global welfare or performance
 - sharing information, synchronizing actions, avoid redundancy, avoid deadlock, livelock.



Techniques of coordination

- ◆ Coordination without communication:
 - Compiled of social laws and conventions or reason by focal points
 - Inferring other agents via observation
 - Partial global distributed/centralized planning
 - Knowledge-transfer protocol -- blackboard
 - Organization-structure
- ◆ Coordination with communication
 - Contracting
 - negotiation approaches (game-theoretic)
 - Market mechanisms



Distributed Problem Solving

- ◆ Task Sharing:
 - Task decomposition (sometimes unnecessary)
 - Task Allocation
 - Task Accomplishment
 - Result Synthesis
- Result Sharing: enhance
 - Confidence
 - Completeness
 - Precision
 - Timeliness



Distributed planning

[Edmond Durfee]:

- ◆ Centralized planning for distributed plans
- ◆ Distributed planning for centralized plans
 - Cooperative planning
 - manufacturing and logistics
- ◆ Distributed planning for distributed plans
- ◆ plan merging, iterative plan formation, negotiation in distributed planning



Centralized Planning for Distributed Plans

- ◆ Given a goal description, a set of operators, and initial state description, generate a partial order plan.
- ◆ Decompose the plan into sub-plans such that ordering relations between steps tend to be concentrated within subplans and minimized across subplans
- ◆ Insert Synchronization (typically communication) actions into subplans.
- ◆ Allocate subplans to agents.
- ◆ Initiate plan execution and monitor progress.
- ◆ Synthesize feedback from agents to ensure complete execution.



Distributed Planning for Centralized Plans

- Task sharing and result sharing
- Task sharing includes task decomposition, task allocation, task accomplishment, result synthesis
- Results sharing improve group performance in terms of confidence, completeness, precision, and timeliness



Post-planning coordination

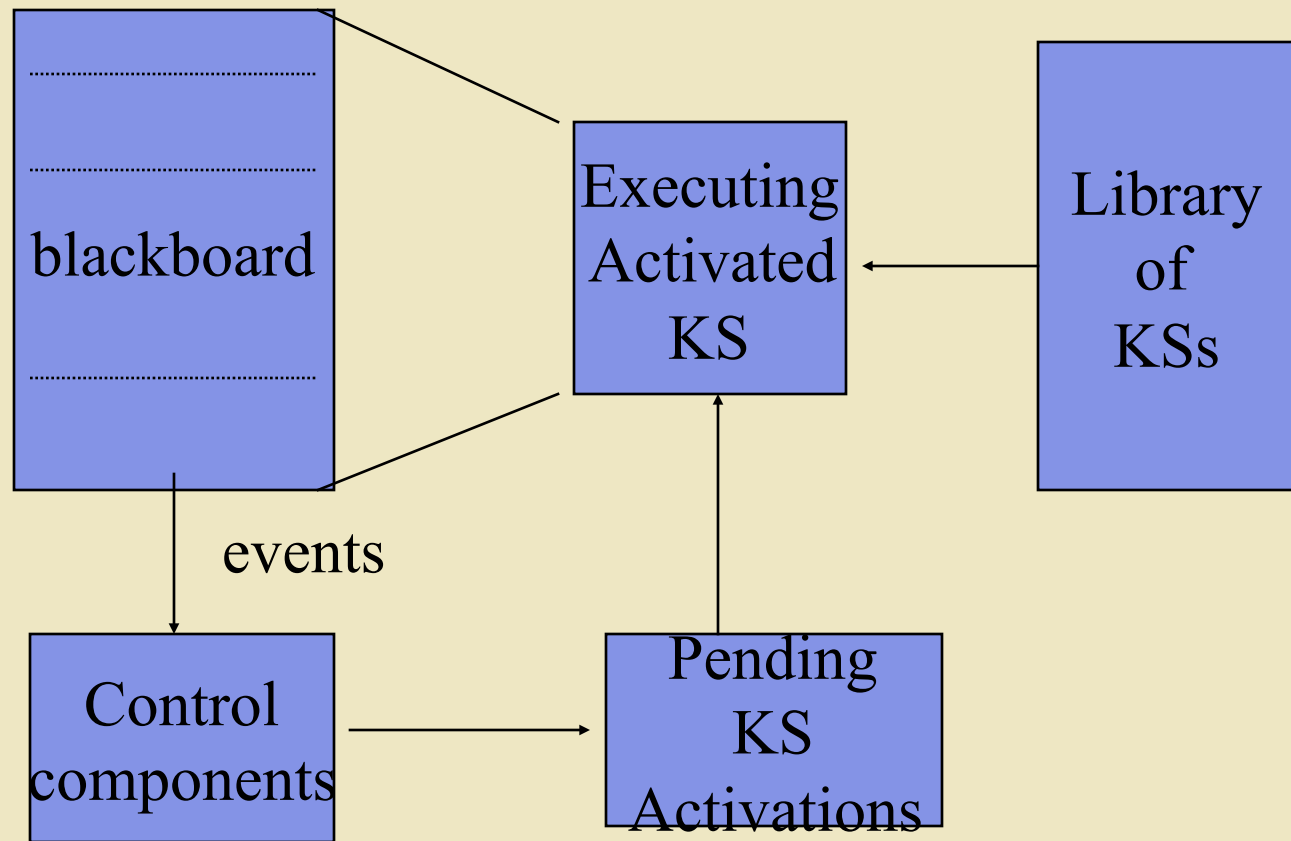
- ◆ Contingency planning:
Retain a lot of alternative plans with choice conditions
- ◆ Monitoring and replanning: Plan repair
 - Organizational structure maybe helpful



Pre-planning coordination

- ◆ Use Social laws: Provide constraints to avoid undesirable states to occur

Blackboard– A cooperative problem solving model





Coordination by organization structure

- ◆ Pre-defined roles, responsibilities and preferences of agents
- ◆ Pre-defined control and communication protocols among agents
- ◆ Prioritizing tasks over agents (allow overlapping responsibilities)
- ◆ Organizational agents are not necessary cooperative; they can be competitive



Coordination by a market mechanism

- ◆ Coordination with a large number of unknown agents
- ◆ Coordination with minimal number of direct communication among agents
- ◆ The market reach competitive equilibrium when
 - 1) consumer bids to maximize their utility, subjected to budget constraints
 - 2) provider bids to maximize their profit, subjected to technology capacity
 - 3) net demand of good is zeroCompetitive equilibrium = Pareto efficient solution



Coordination of tasks

- ◆ Decomposition of tasks
- ◆ Distribution of tasks
- ◆ Control or coordination
 - Determine shared goal
 - Determine common tasks
 - Avoid unnecessary conflicts
 - Pool knowledge and evidence



Task decomposition

- ◆ Divide and conquer
 - AND/OR tree
 - Spatial decomposition vs functional decomposition
- ◆ Depends on designer's choice
- ◆ Must consider resources and capabilities of agents



Distribution of tasks

- ◆ Market mechanisms: generalized agreement and mutual selection
- ◆ Contract net
- ◆ Multi-agent planning
- ◆ Organizational structure
- ◆ Recursive allocations
- ◆ Agent-mediated matchmaking/brokerage



Learning to coordinate

- ◆ Learning coordinate actions [Gerhard Weiß]
 - Collective learning
 - ACE algorithm (action estimation)
 - AGE algorithm (action group estimation)
- ◆ Learning multi-agent reinforcement [Ming Tan]:
 - Sharing sensory information
 - Sharing learned policies,
 - Learning joint tasks

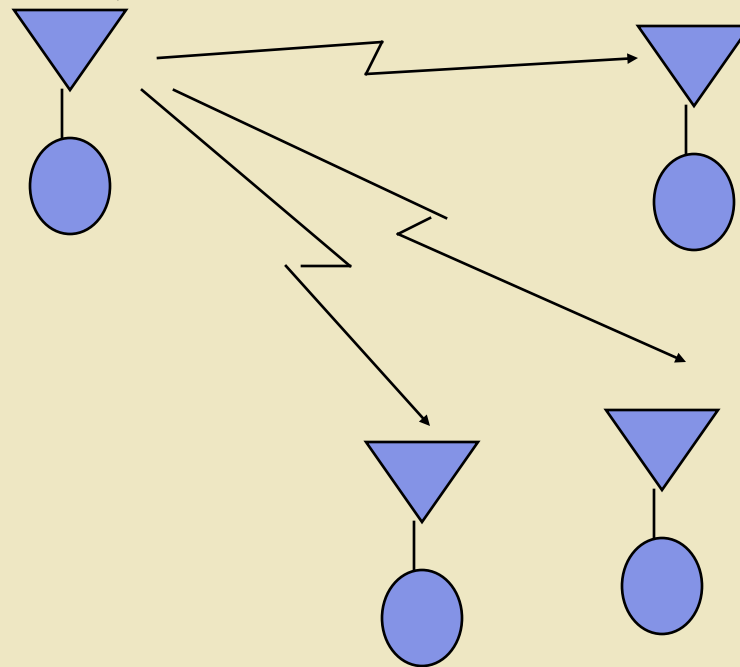


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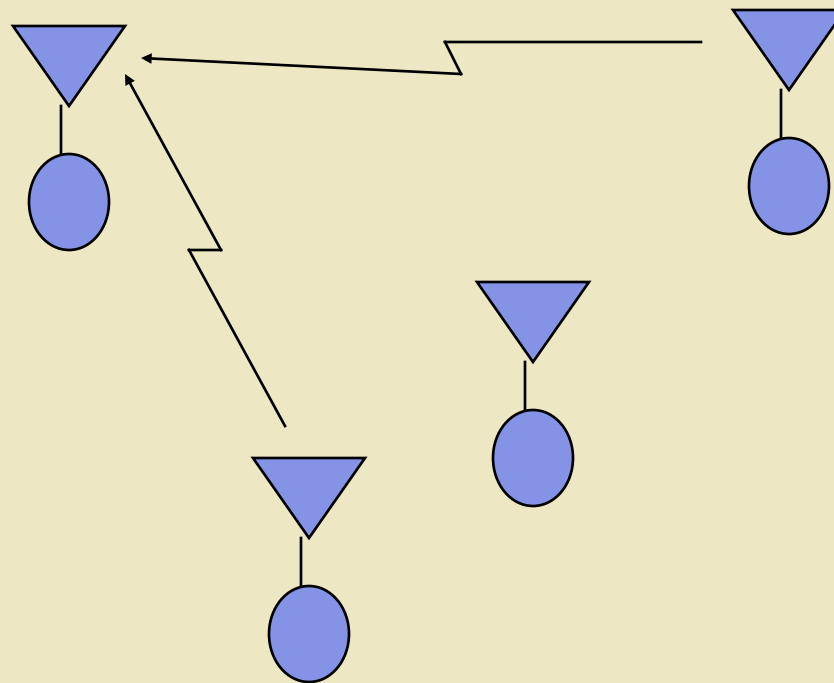
Contract net protocol

- ◆ Manager announces tasks via (possible selective) multicast



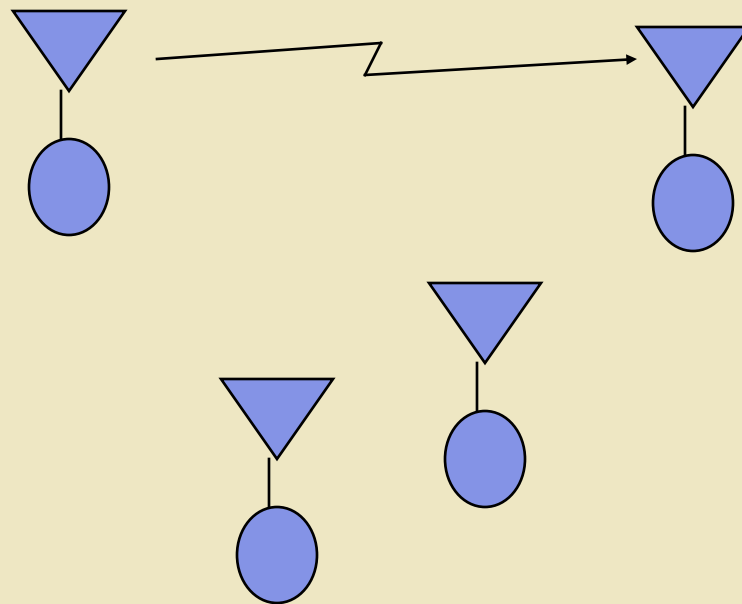
Contract net protocol

- ◆ Agents evaluate the announcement, Some of the agents submit bids



Contract net protocol

- ◆ The manager awards a contract to the most appropriate agent





Task announcement

- ◆ Eligibility specification: criteria that a node must meet to be eligible to submit a bid
- ◆ Task abstraction: a brief description of the task to be executed
- ◆ Bid specification: a description of the expected format of the bid
- ◆ Expiration time: a statement of the time interval during which the task announcement is valid



Bid and award messages

- ◆ A bid consists of a node abstraction – a brief specification of the agent's capabilities that are relevant to the task
- ◆ An award consists of a task specification – the complete specification of the task



Applicability of contract net

- ◆ The contract net is
 - A high level communication protocol
 - A way of distributing tasks
 - A means of self-organization for a group of agents
- ◆ Best used when
 - The application has a well defined hierarchy of tasks
 - The problem as a coarse-grained decomposition
 - The subtasks minimally interact with each other, but cooperate when they do



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Rosenschein's Work on Rules of Encounter

- ◆ Negotiation on different domains
 - Task oriented domain (postmen, database, fax)
 - State oriented domain (block world)
 - Worth oriented domain (agents rank the worth on different states)
 - Information oriented domain (information sharing)



State-oriented domain

- ◆ $\text{SOD}\langle S, A, J, c \rangle$
 - S: states
 - A: agents
 - J: joint plans
 - c: cost function of agent's role in a joint plan



Worth oriented domain

◆ WOD<S,A,J,c>

- S: states
- A: agents
- J: joint plans
- C: cost function of agent's role in a joint plan
- W: mapping worth function of a given state for a given agent

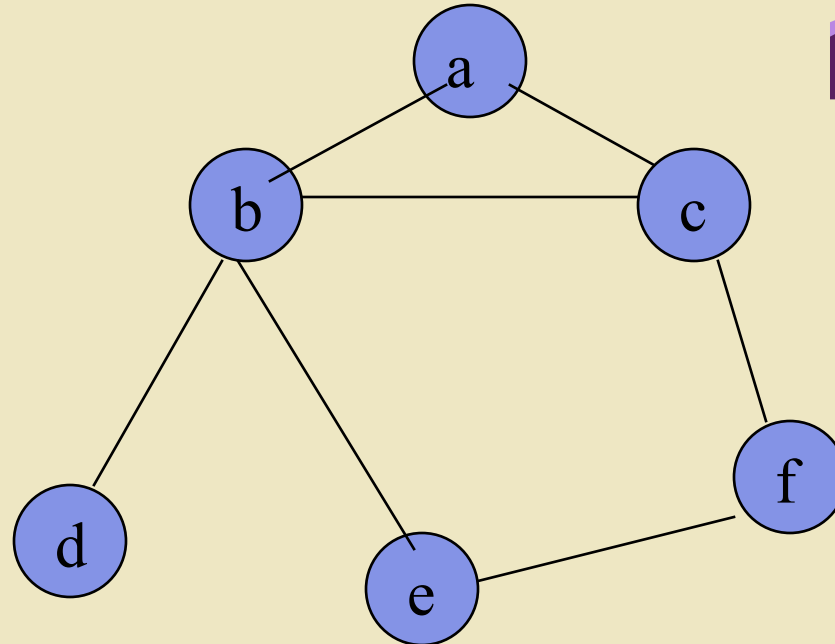


Task oriented domain

- ◆ $TOD\langle T, A, c \rangle$
 - T: tasks
 - A: agents
 - c: cost function of executing tasks by an agent
- ◆ Original set of tasks $\{T_1, T_2\}$
- ◆ Negotiated set of tasks $\{D_1, D_2\}$
- ◆ Utility of a deal δ for agent i is
$$utility_i(\delta) = cost(T_i) - cost(D_i)$$

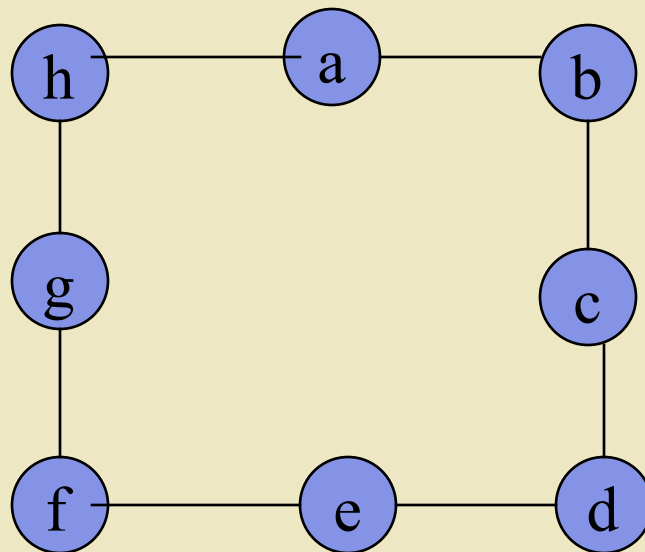
Postmen Domain

Post office



Deception in negotiation— postmen domain

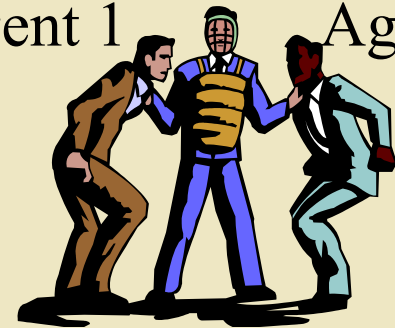
Post office



True task

Assignment:
Must return to office

Agent 1 Agent 2

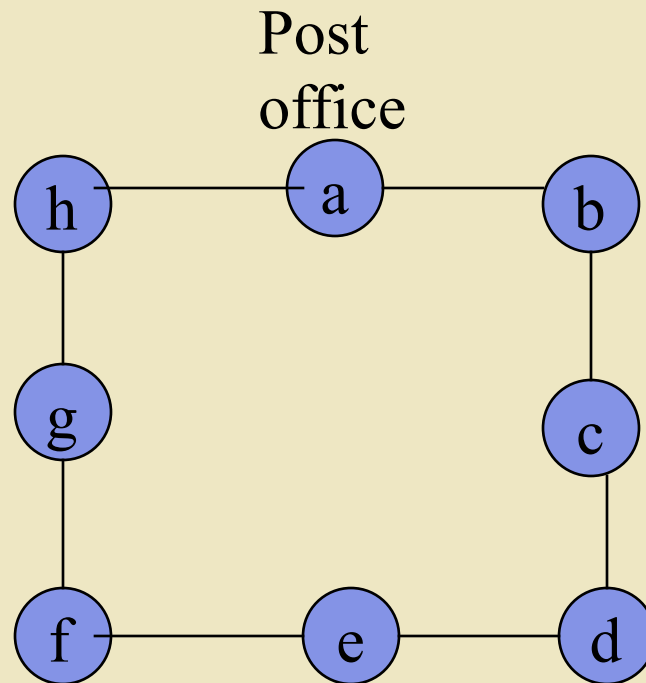


$\{b, f\}$

$\{e\}$

Flip a coin to decide who
Deliver all the mails

Hiding task



Task claim during negotiation

Agent 1 Agent 2



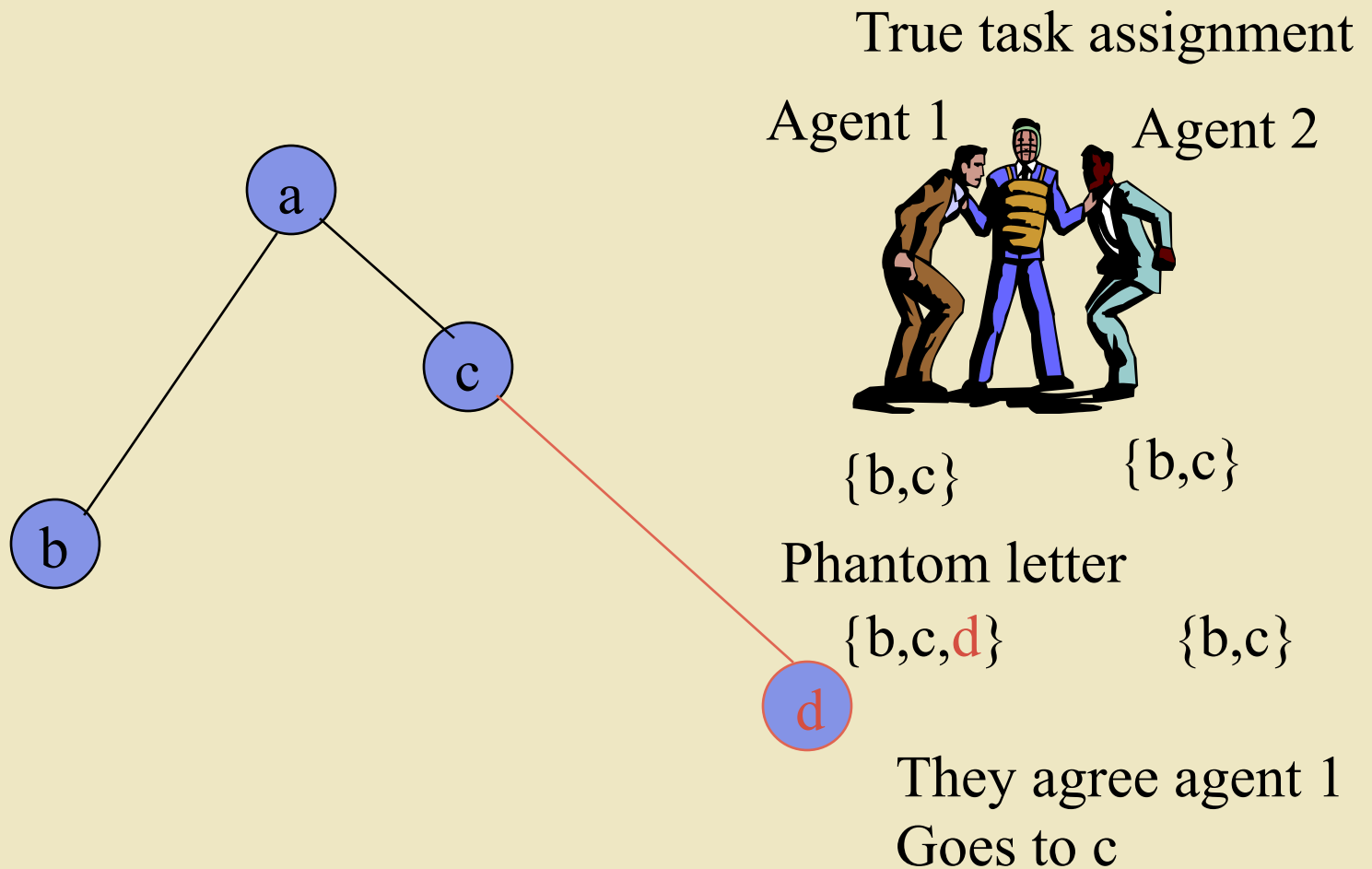
$\{f\}$

$\{e\}$

$\{b\}$

They decide agent 2
Delivers all the mails

Phantom task

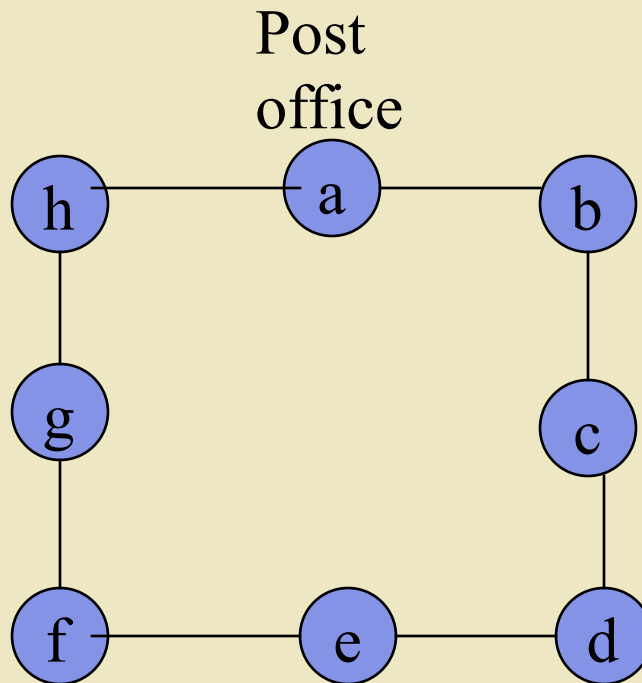




Negotiation over mixed deals

- ◆ Divide the tasks $\{D1, D2\}$
- ◆ Agent 1 has probability p to take $D1$ and $1-p$ to take $D2$
- ◆ Agent 2 has probability $1-p$ to take $D1$ and p to take $D2$
- ◆ Change discrete deals to continuous deals

Hiding letters with mixed all-or-nothing deals



Task claim during negotiation

Agent 1 Agent 2



$\{f\}$

$\{e\}$

$\{b\}$

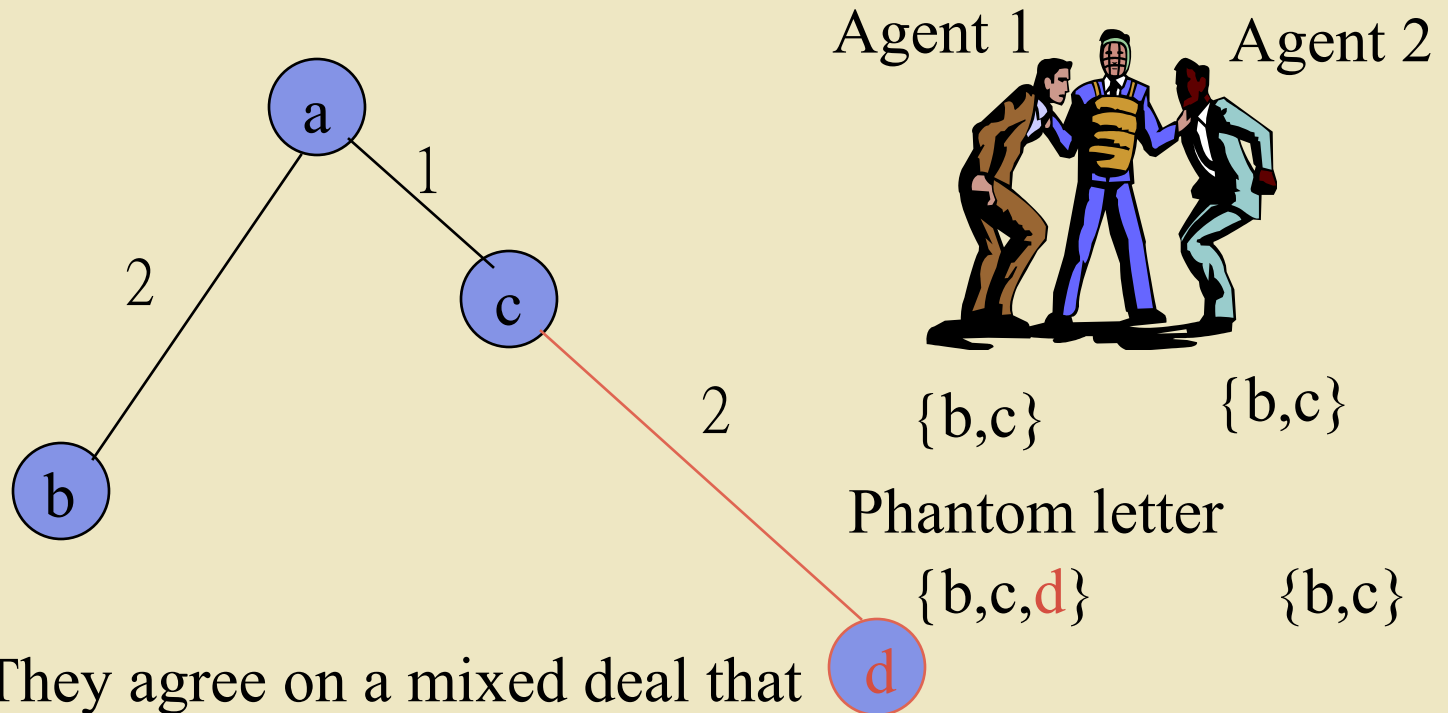
They will agree on the mixed deal where agent 1 has 3/7 chance of delivering to f and e



Mixed deal prevents hiding tasks

- The expected utility of agent 1 with honest bid is $\frac{1}{2} * 8 = 4$
- Agent 1 might still have a chance to delivery all letters even if he hide the delivery task b
 - The expected utility of agent 1 with deception is $(6/14) * 8 + (8/14) * 2 = 64/14 = 4.57$, there is no reason to hide the task under the mixed all-or-nothing deal

Phantom letters with Mixed deals



They agree on a mixed deal that
Agent 1 has $5/8$ to deliver all letters



Mixed deal prevents phantom tasks

- ◆ The agent 1 with honest bid has expected utility $\frac{1}{2} * 6 = 3$
- ◆ With phantom letter, the agent 1 has the expected utility $\frac{5}{8} * 6 = 3.75$
- ◆ no reason to propose a phantom letter



Incentive compatibility mechanism

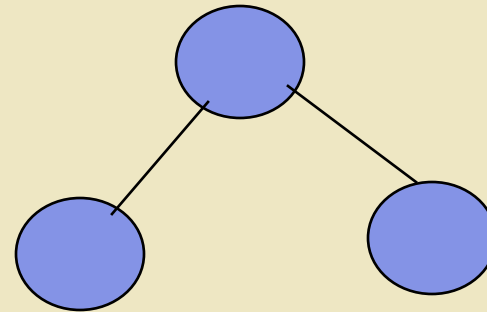
◆ Theorem

For all encounters in all sub-additive TODs, when using a PMM(product-maximizing mechanism) over all-or-nothing deals, no agent has an incentive to hide a task.

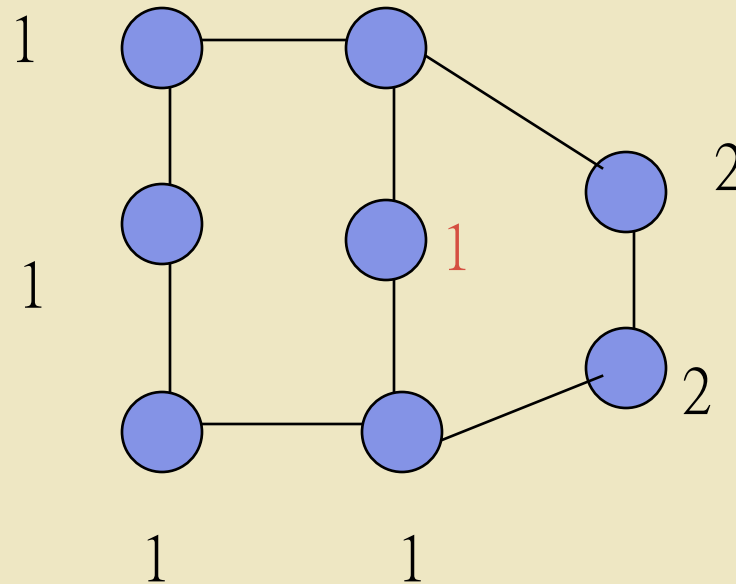
Sub-additive TOD

- ◆ $c(X \cup Y) \leq c(X) + c(Y)$
- ◆ Postmen (returning to post office)
- ◆ Database domain
- ◆ Fax domain

Post office



Decoy tasks



Agent 1 can declare the fake task **1** to lock itself at the original path otherwise it will have to take agent 2's tasks



Concave TOD

- ◆ Concave TOD is a subset of sub-additive TOD's that satisfy:
If X, Y are set of tasks, X is subset of Y
then adding other set of tasks Z ,
$$c(X \cup Z) - c(X) \geq c(Y \cup Z) - c(Y)$$
- ◆ The fax, database and postmen (restricted to trees) are concave



Modular TOD

- ◆ $c(X \cup Y) = c(X) + c(Y) - c(X \cap Y)$
- ◆ Any modular TOD is also concave and sub-additive
- ◆ Fax domain is modular TOD, database and postmen domain (unless star topology) are not.



Incentive compatibility table

T: true telling L: lying

T/P: true telling, lying might sometimes be beneficial
but can be caught with high penalty

Sub-additive

concave

modular

	H	P	D
pure	L	L	L
A/N	T	T/P	L
Mix	L	T/P	L

	H	P	D
pure	L	L	L
A/N	T	T/P	L
Mix	L	T/P	L

	H	P	D
pure	L	L	L
A/N	T	T/P	L
Mix	L	T/P	L



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Why game theory?

- ◆ Provide fundamental explanation of multi-agent decision making behavior on various situations
- ◆ Previous work showed that [Rosenschein and Genesereth, 1985; Rosenschein, 1994, Haynes and Sen, 1996, Wu and Soo, 1998a, 1999].



Underlying assumptions

- ◆ Rational agent assumption
 - maximize its own expected utility (selfish)
 - Mutual rationality



Previous results of game theory

- ◆ On the rationality assumption of agents, agents will try to reach a stable Nash equilibrium
- ◆ All rational agents will not leave the Nash equilibrium they have reached
- ◆ Rational agents are able to coordinate and cooperate with a game theoretical deal-making mechanism



Basic notions of game theory

- ◆ Dominant strategy
- ◆ Nash equilibrium state
- ◆ Pareto efficient/optimal state
- ◆ Pareto dominant strategy



Dominant strategy

- ◆ The strategy S_i^* is a dominant strategy if it is player i 's strictly best response to player $-i$'s strategy S_{-i} :



Nash equilibrium

- ◆ A strategy combination (s_i^*, s_{-i}^*) is a Nash equilibrium if any agent will get less its payoff when it deviates from this strategy combination alone.
- ◆ Dominant strategies' combination state will lead to a Nash equilibrium but not vice versa



Pareto-efficient

- ◆ A strategy combination is Pareto efficient if no other strategy combination increases the payoff of one agent without decreasing the payoff of another agent.



Pareto-dominant

- ◆ If a strategy combination X is strongly Pareto-dominates another strategy combination Y ,
- ◆ then all agents have higher payoffs at X .



What are difficult games?

- Prisoner's dilemma [A Nash equilibrium that is not Pareto-efficient]
- Games with no Nash equilibrium
- Games with multiple Nash equilibria



Negotiation mechanisms in difficult games

- In Agent'99, Wu and Soo showed an agent negotiation mechanism that can get around these difficulty game situations using guarantee and compensation communication actions involving with a trusted third party



A Prisoner's dilemma game

- ◆ A game that has a unique Nash equilibrium but the Nash equilibrium is Pareto-dominated by some other strategy combination.

Prisoner's dilemma and dilemma-free games



		A	
		cooperate	defect
B	cooperate	1, 1	-2, 2
	defect	2, -2	-1, -1

Diagram (a) illustrates a Prisoner's Dilemma. The payoff matrix shows that for Player A, defecting is a dominant strategy (2 > 1 and -1 > -2), and for Player B, defecting is also a dominant strategy (-2 > 1 and -3 > -1). The outcome (defect, defect) with payoffs (-1, -1) is circled, representing the Nash equilibrium, which is Pareto inefficient.

(a)

		A	
		cooperate	defect
B	cooperate	1, 1	0, -2
	defect	0, -2	-3, -3

Diagram (b) illustrates a dilemma-free game. The payoff matrix shows that for Player A, cooperating is a dominant strategy (1 > 0 and -2 > -3), and for Player B, cooperating is also a dominant strategy (1 > 0 and -2 > -3). The outcome (cooperate, cooperate) with payoffs (1, 1) is circled, representing the Nash equilibrium, which is Pareto efficient.

(b)

A prisoner's dilemma game

		Q	
		Cooperate	Fink
P	Cooperate	-1	-10
	Fink	0	-8

Diagram (a) illustrates a Prisoner's Dilemma game matrix. The matrix shows payoffs for two players, P and Q, who can choose to Cooperate or Fink. The payoffs are: (Cooperate, Cooperate) = (-1, -1), (Cooperate, Fink) = (-10, 0), (Fink, Cooperate) = (0, -10), and (Fink, Fink) = (-8, -8). A blue oval highlights the (Fink, Fink) outcome, indicating it is the Nash equilibrium. Arrows indicate that for each player, Finking is the dominant strategy.

(a)

		Q	
		Cooperate	Fink
P	Cooperate	-1	-10
	Fink	-1.1	-9.1

Diagram (b) illustrates a dilemma-free game matrix. The matrix shows payoffs for two players, P and Q, who can choose to Cooperate or Fink. The payoffs are: (Cooperate, Cooperate) = (-1, -1), (Cooperate, Fink) = (-10, -1.1), (Fink, Cooperate) = (-1.1, -10), and (Fink, Fink) = (-9.1, -9.1). Blue ovals highlight the (Cooperate, Cooperate) and (Fink, Fink) outcomes, indicating they are the Nash equilibria. Arrows indicate that for each player, Cooperating is the dominant strategy.

(b)

Prisoner's Dilemma game matrix (a) A special case of a PD game matrix. (b) A dilemma-free game matrix. Note: (-1,-1) Pareto-dominate all three other strategy combinations



Create an Equilibrium by Negotiation

- ◆ We propose two communication actions in the negotiation process
 - Ask guarantee; offer guarantee
 - Ask compensation; Offer compensation



The need of a trusted third party

- ◆ Traditional game theory cannot deal with the games without Nash equilibrium or more than one Nash equilibrium
- ◆ It cannot deal with prisoner's dilemma too
- ◆ We propose to use a trusted third party to remedy the drawback



Roles of a trusted third party

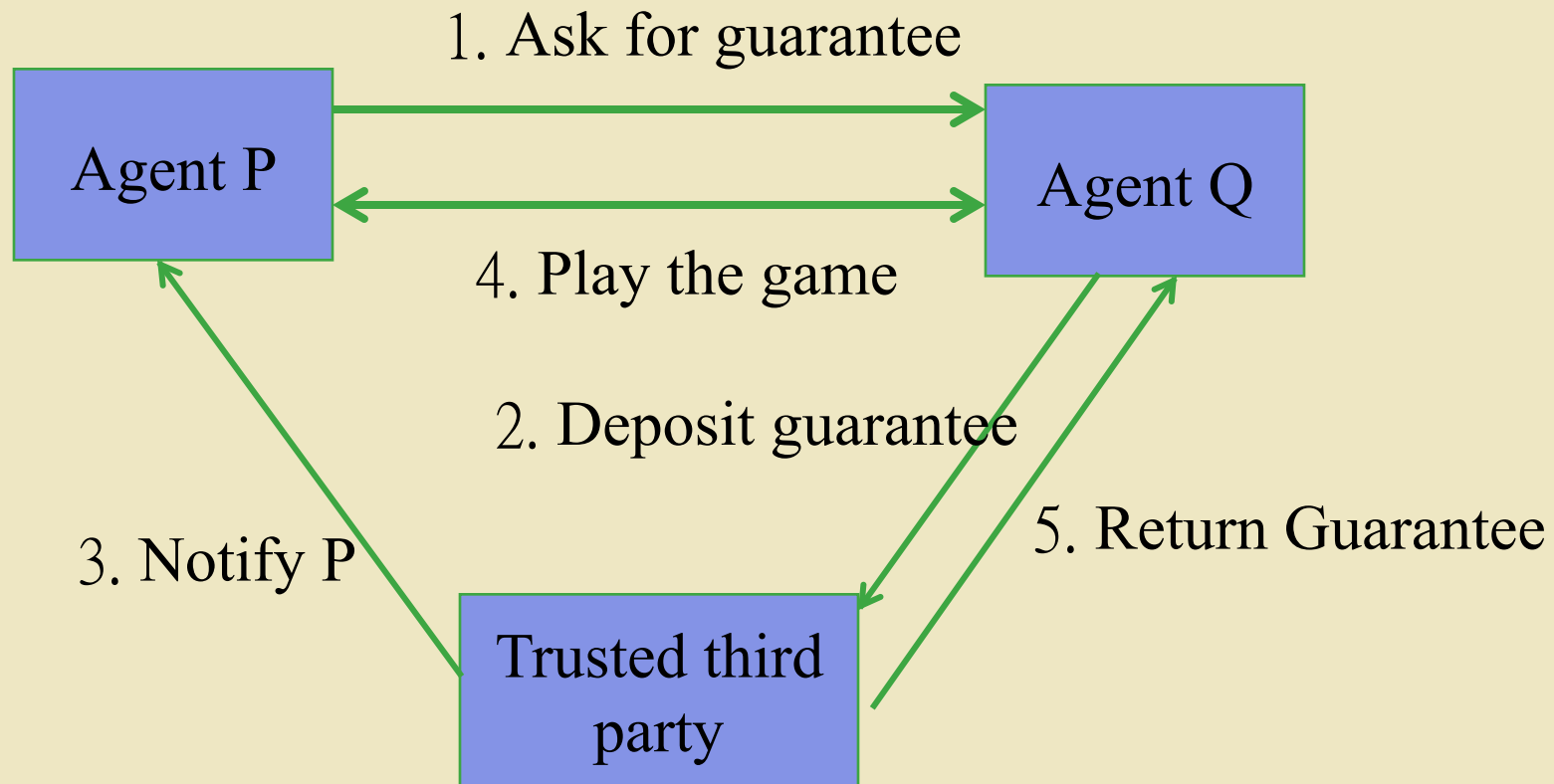
- ◆ Examples: bank; government; court; referee
- ◆ an intermediary agent
 - temporary holding the deposit of guarantee or compensation from one agent
 - forfeit or return the guarantee or compensation deposited if the other side does not obey the agreement



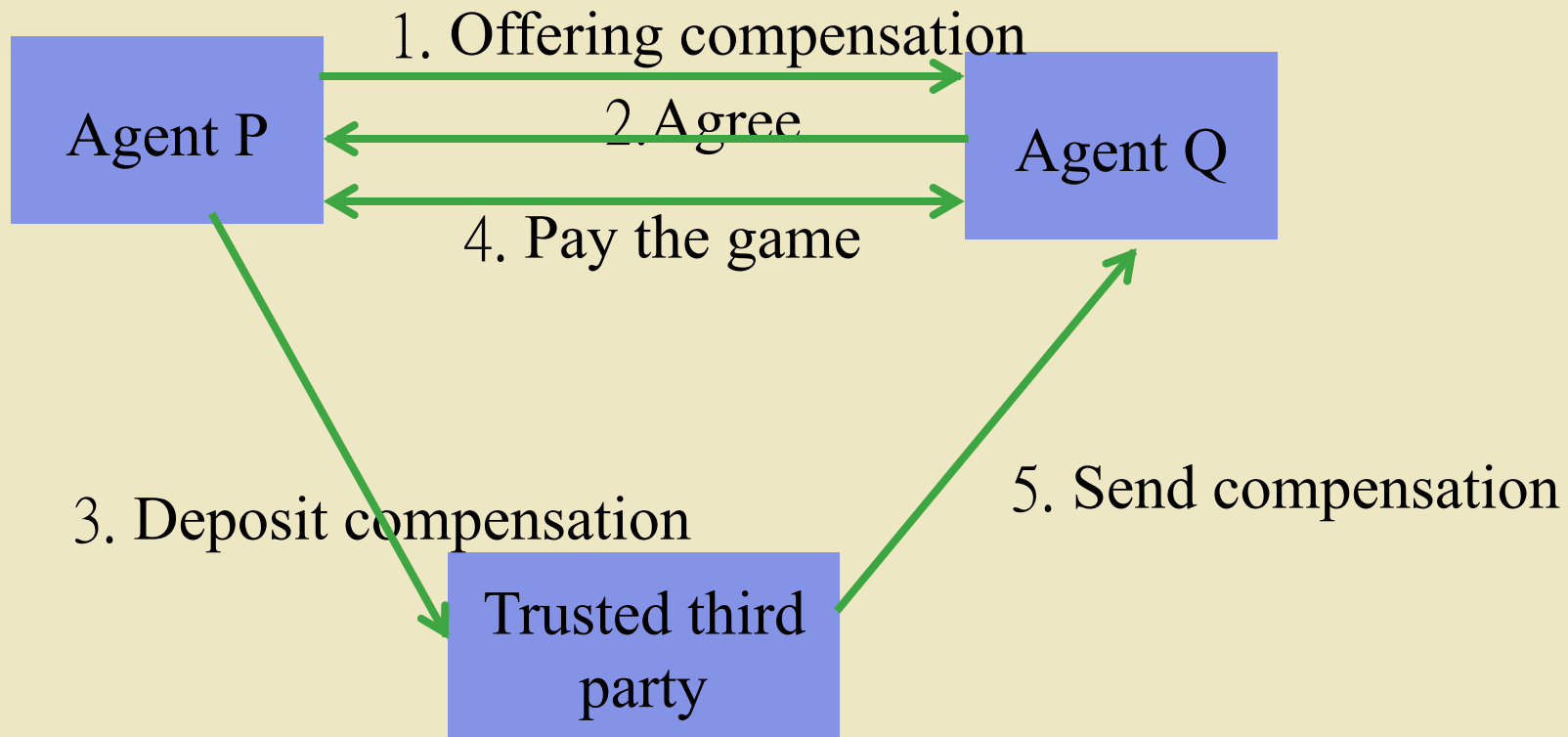
TTP negotiation mechanism

- ◆ A trusted third party exists to enforce the commitments and contracts negotiated by both agents
- ◆ Allows negotiating agents to use compensation and guarantee communication actions in negotiation

The guarantee communication action



The compensation communication action



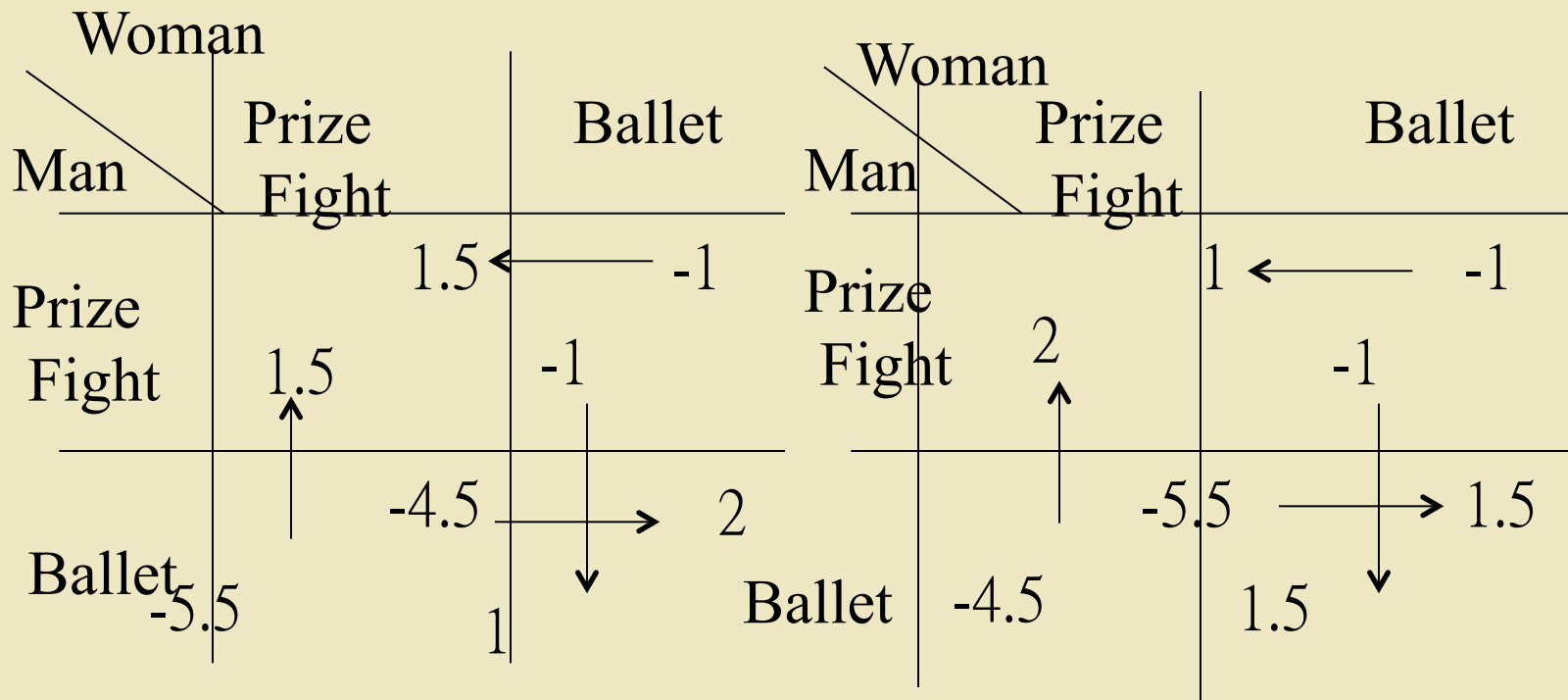
Battle of Sexes Game with Multiple Nash Equilibria



Woman		Prize Fight	Ballet
Man	Prize Fight	1, 2	-1, -1
	Ballet	-5, -5	2, 1

After Negotiation with communication actions

Man offering compensation 0.5 for Woman to play "Prize Fight" & Woman offering 0.5 compensation for Man to play "Ballet".



A Welfare Game without Nash Equilibrium

G \ P	Work	Idle
	2, 3	3, 1
Aid	3, -1	-1, 0
No Aid	-1, 1	0, 0

(a)

G \ P	Work	Idle
	2, 3	1, 1
Aid	3, -1	-1, 0
No Aid	-1, 1	0, -2

(b)

A Welfare Game (a) The original game matrix. (b) The game matrix after negotiation.



The general negotiation procedure

- Getting all payoff entries and constructing the game matrix.
- Reasoning on the game matrix, finding the sets
 - 2.1. The Nash equilibrium set N .
 - 2.2. The Pareto-efficient set P .
- Loop of offer, counter offer, and negotiation.
 - 3.1. agent i offers a particular strategy combination.
 - 3.2. agent $-i$ accepts the offer or proposes another counter offer.
[based on certain concession strategy]
 - 3.3 Negotiation ends on a compromised outcome or no more offers exist.



Proper quantum principle

- ◆ Since the basic quantum of payoff may not exist in general cases.
- ◆ The compensation should not be less than the amount that another agent offered.
- ◆ This principle is necessary to prevent that one agent may offer so small compensation that causes a lengthy negotiation process.



Equal concession principle

- ◆ Compensation negotiation
 - Each agent must have equal amount of compensation offer at each step of offer
 - (This is an idealized assumption)



Zeuthen Strategy

- ◆ Negotiation Agent who has less risk will make concession.

Risk_i is the willingness of agent i to risk

- ◆ Risk_i^t = 1 if utility_i(D_i^t) = 0 ; Otherwise

Risk_i^t = [utility_i(D_i^t) – utility_i(D_j^t)] / utility_i(D_i^t)

[D_j is to accept agent j's proposal]

- ◆ Make no concession if your Risk is greater than the other agent



Effect of Zeuthen strategy

- ◆ When both agents use monotonic concession protocol and the Zeuthen, it will lead agents to agree on the deal in the negotiation set. The set of all conflict free deals will be individual rational and pareto optimal one and the conflict deal will maximize the Nash product.
- ◆ https://en.wikipedia.org/wiki/Zeuthen_strategy



Possible Multilateral Concession Criteria

- ◆ (1) Strong concession: Make a proposal that is strictly better for each of the other agents.
- ◆ (2) Weak concession: Make a proposal that is strictly better for at least one of the other agents.
- ◆ (3) Pareto concession: Make a proposal that is no worse for the other agents and strictly better for one of them.



Cont'd

- ◆ (4) Utilitarian concession: Make a proposal such that the sum of utilities of the other agents increases.
- ◆ (5) Egalitarian concession: Make a proposal such that the minimum utility amongst the other agents increases.
- ◆ (6) Nash concession: Make a proposal such that the product of utilities of the other agents increases.
- ◆ (7) Egocentric concession: Make a proposal that is worse for yourself.



Does the negotiation protocol terminate?

- ◆ Will negotiation protocol lead to live-lock or dead-lock?
- ◆ Eg. Cyclic or no valid concession.
- ◆ <https://staff.fnwi.uva.nl/u.endriss/slides/contributed-talks/endriss-aamas-2006.pdf>



Homework 2 Due April 10

- ◆ Read pdf file
<https://staff.fnwi.uva.nl/u.endriss/slides/contributed-talks/endriss-aamas-2006.pdf>
- ◆ Show that proposition 2, multi-lateral negotiation protocols by using different concession strategies, some may lead to dead-lock free, some may not.



Theorem (Existence of Nash and Pareto Equilibrium)

- ◆ In one-shot negotiation game, with the asking guarantee and offering compensation mechanisms, there exists at least one Nash Equilibrium which Pareto weakly dominates all other outcomes.



Theorem (Order independent)

- ◆ In one-shot negotiation game, with the asking guarantee and offering compensation mechanisms, the final negotiated utility is independent of the order of the initiating the negotiation.
- ◆ All the resulting Nash Equilibrium outcomes which Pareto weakly dominate all other outcomes will end up with the same utility values.



Theorem (Convergence of the negotiation procedure)

- ◆ In one-shot negotiation game, with the offering compensation mechanisms, the negotiation procedure will end in finite time.



Negotiation without knowing other agent's payoff

- ◆ Previous game theoretic reasoning of a Nash equilibrium must assume complete payoff matrix
- ◆ In real world applications, however, it is often difficult to obtain other agent's payoff information.
- ◆ How can agents negotiate without knowing other agent's payoffs?



Min-max strategy may not be Pareto-optimal

- ◆ Conservative agents using min-max strategies to the game with incomplete information may lead to sub-optimal negotiation results.



What is an NFD equilibrium?

- ◆ Traditional game theory
- ◆ Nash equilibrium: A stable state where no agent will deviate alone by gaining more payoff.
- ◆ No-Fear-of-Deviation (NFD) equilibrium: a stable state where even other agents deviate, no single agent will get less payoff.



Q \ P	P1	P2
Q1	2	-2
Q2	-1	1

A complete payoff matrix

P's view

Q \ P	P1	P2
Q1	2	-2
Q2	-1	1

Q's view

Q \ P	P1	P2
Q1	?	?
Q2	?	?

Min-max results in incomplete information

Q \ P	P1	P2
Q1	2	-2
Q2	-1	1

P will choose P1

Q \ P	P1	P2
Q1	?	?
Q2	?	?

Q will choose Q2

Without communication, P and Q using min-max will end up with (Q2, P1) which is not a Pareto-optimal state



Q asks P to pay guarantee

Q \ P	P1	P2
Q1	? 2	? -4 -2 +4
Q2	? -1	? -4 1 +4

Q asks P to pay guarantee 4 to play P1



P asks Q to pay guarantee

Q ^P	P1	P2
Q1	2 ?	-2 ?
Q2	-1+3 ?-3	1+3 ?-3

P asks Q to pay guarantee 3 to play Q1

Final results

Q \ P		P	
		P1	P2
Q	Q1	2 2	-2-4 -2+4
	Q2	-1-3 -1+3	1-4+3 1+4-3

Agreed at the created NFD (P1,Q1),

P pays guarantee 4

Q pays guarantee 3

A Prisoner's dilemma game

Q \ P		P	
		P1	P2
Q	Q1	-1, -1	0, -10
	Q2	-10, 0	-8, -8

Complete information:

Created Nash (P1, Q1)

Q asks P to deposit guarantee 1 to play P1

P asks Q to deposit guarantee 1 to play Q1

TTP negotiation under incomplete payoff information

Q \ P	P1	P2
Q1	-1	0
Q2	-10	-8

Reach created NFD equilibrium

P asks Q to pay guarantee
9 to play Q1

Q asks P to pay guarantee
9 to play P1

Battle of Sexes –

A game with multiple Nash

Q \ P	P1	P2
Q1	1 3	-1 -1
Q2	-5 -5	-1 1

Handwritten annotations on the table:

- A dashed arrow points from the (Q1, P1) cell to the (Q2, P2) cell.
- A red circle highlights the (Q2, P2) cell.
- Green numbers are added to the (Q2, P1) and (Q2, P2) cells:
 - In (Q2, P1), -5 is changed to $-5+1$ and -1 is changed to $-5-1$.
 - In (Q2, P2), -1 is changed to $3-1$ and 1 is changed to $1+1$.

Complete information:

Agree at the created
Nash (Q2,P2)

P pays compensation
1 to Q to play Q2

Battle of Sexes— A game with multiple Nash

Q \ P		P1	P2
		P1	P2
Q	Q1	1-6 3-3	-1+3 -1-3
	Q2	-5-1 -5+1	-6 3-1 1-1

Incomplete information:

P pays compensation 1 to
Q to play Q2

Q asks P to pay guarantee 6
To play P2

P asks Q to pay guarantee 3
To play Q2

Agree at created NFD
Equilibrium (Q2,P2)

Social welfare— A game with no Nash

		P	
Q		P1	P2
Q	Q1	0.5 $2 - 0.5$ $4 + 0.5$	1
	Q2	-1 5	0

Complete information:

P pays compensation 0.5 to
Ask Q to play Q1

P asks Q to pay guarantee
1 To play Q1

Agreed at created Nash (Q1,P1)

Social welfare— A game with no Nash

		P	
		P1	P2
Q	Q1	0.5 $2-0.5$ $4+0.5$ 3	$1-3$ $1+3$
	Q2	$-1+3$ $5-3$	$0+3-3$ $0-3+3$

incomplete information

P pays compensation 0.5 to
Ask P Q to play Q1

Q asks P to pay guarantee
3 to play P1

P asks Q to pay guarantee
3 to play Q1

Agree at created NFD (Q1,P1)

Summary

	Prisoner's dilemma games	Battle of Sexes Games	Social Welfare Games
Traditional game theory without communication	Trap into sub-optimal Nash Mini-max doesn't help	Cannot predict result with pure strategy Mini-max strategy may lead to suboptimal solution	No Nash equilibrium with pure strategy Mini-max strategy may lead to suboptimal solution
Allows TTP negotiation Complete Payoff information	Created Pareto-optimal Nash Q pays guarantee 1 P pays guarantee 1	Created Pareto-optimal Nash P pays compensation 1	Created Pareto-optimal Nash P pays compensation 0.5 Q pays guarantee 1
Allows TTP negotiation incomplete Payoff information	Created Pareto optimal NFD Q pays guarantee 9 P pays guarantee 9	Created Pareto-optimal NFD P pays compensation 1 and pays guarantee 6 Q pays guarantee 3	Created Pareto-optimal NFD P pays compensation 0.5 and pays guarantee 3 Q pays guarantee 3



Negotiation under uncertainty and risk

- ◆ Key motivations and arguments:
 - How do game theoretic agents negotiate without knowing other agents's exact utility functions and wealth levels in uncertain payoff games?
 - The utility functions are very sensitive to agent's wealth levels
 - To obtain the utility functions and wealth levels about other agents are impractical in real world
 - Monetary payoff (public, tradable, same scale)

VS

Utility payoff (private, non-tradable, different scales)



Underlying assumptions on uncertain games

- ◆ Rational and mutually rational
- ◆ A monetary payoff game matrix is available
- ◆ Agents do not know other's risk preference, wealth levels and utility functions
- ◆ Agents can negotiate using compensation and guarantee actions
- ◆ A trusted third party exists to enforce payment of guarantee and compensation payoffs.



An uncertain game in probabilistic monetary payoffs

		P	
		C	D
Q	A	0/0.5, 40/0.5 5/0.8, 40/0.2	10/0.5, 20/0.5 0/0.33, 30/0.67
	B	15/0.4, 30/0.6 10/0.1, 20/0.9	15/0.4, 40/0.6 0/0.75, 40/0.25



Definition of risk preference

Risk averse:

$$\int u(x) dF(x) < u\left(\int x dF(x)\right)$$

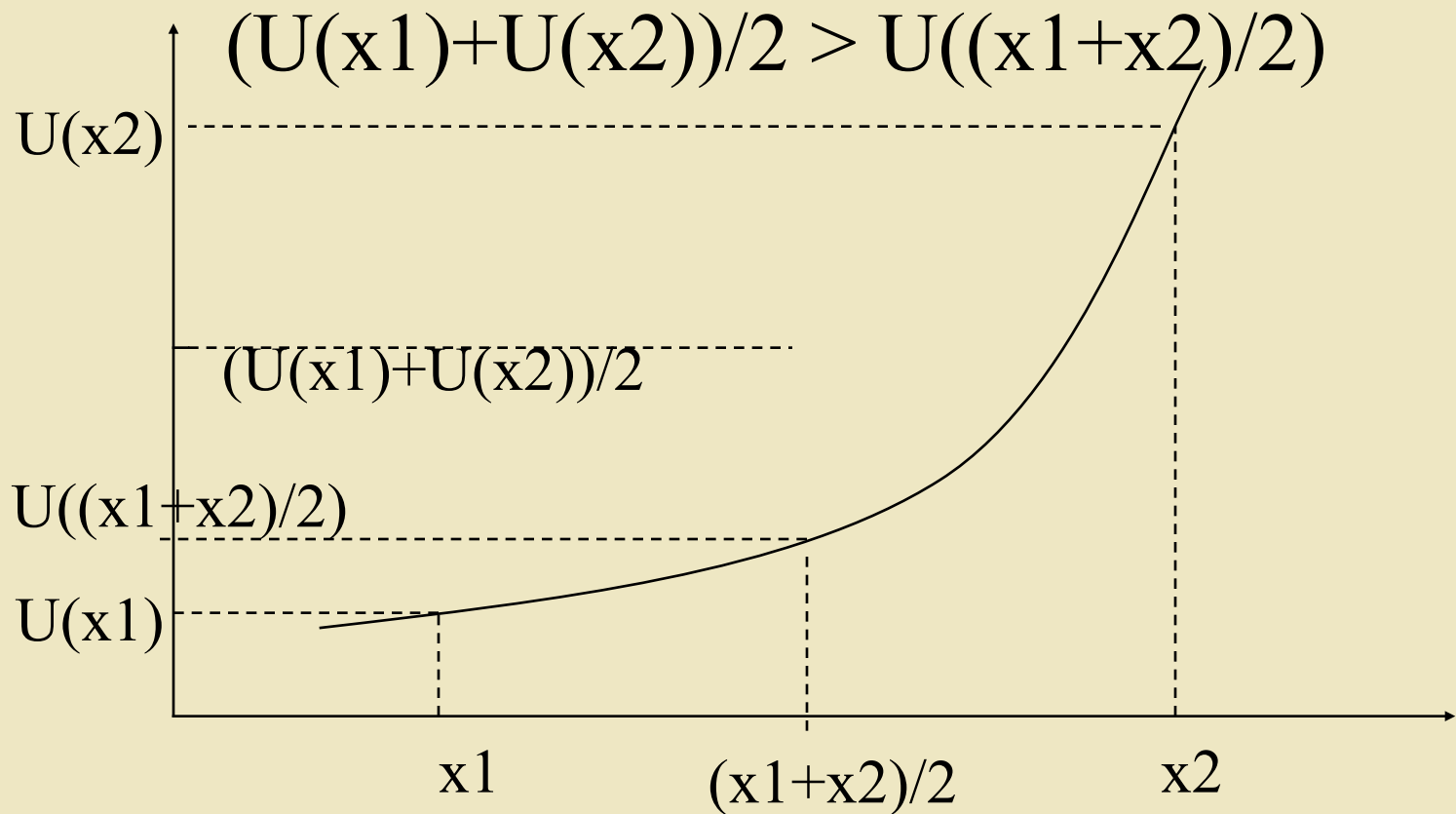
Risk neutral:

$$\int u(x) dF(x) = u\left(\int x dF(x)\right)$$

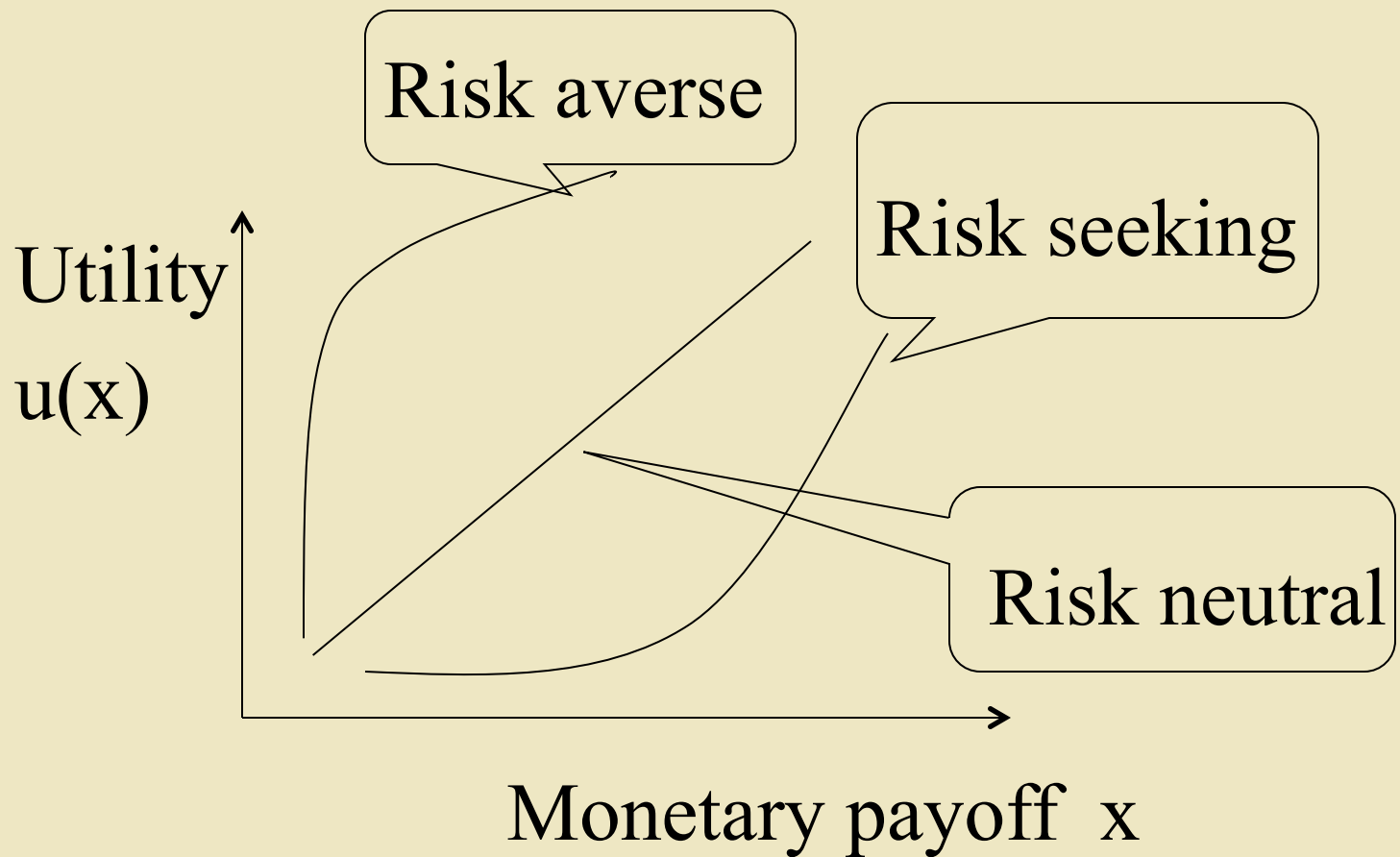
Risk seeking:

$$\int u(x) dF(x) > u\left(\int x dF(x)\right)$$

Risk Seeking example



Risk Preference Types





An uncertain game in probabilistic monetary payoffs

		P	
		C	D
Q	A	0/0.5, 40/0.5 5/0.8, 40/0.2	10/0.5, 20/0.5 0/0.33, 30/0.67
	B	15/0.4, 30/0.6 10/0.1, 20/0.9	15/0.4, 40/0.6 0/0.75, 40/0.25



Calculation of expected utility from monetary payoffs

	Expected utility for P	Expected utility for Q
(A,C)	$.5*u(w+0)+.5*u(w+40)$	$.8*u(w+5)+.2*u(w+40)$
(A,D)	$.5*u(w+10)+.5*u(w+20)$	$.33*u(w+0)+.67*u(w+30)$
(B,C)	$.4*u(w+15)+.6*u(w+30)$	$.1*u(w+10)+.9*u(w+20)$
(B,D)	$.4*u(w+15)+.6*u(w+40)$	$.75*u(w+0)+.25*u(w+40)$



payoff matrix in terms of expected monetary payoffs

$\begin{array}{c} P \\ \diagdown \\ Q \end{array}$		C	D
A	20	12	15
B	24	19	30
		10	20



Utility functions of different risk attitudes

- ◆ Risk neutral $u(x)=x$
- ◆ Risk averse $u(x)=\sqrt{x}$
- ◆ Risk seeking $u(x)=x^2$



How agents negotiate under uncertainty?

- ◆ Two-stage negotiation protocol:
 - Compensation negotiation stage
 - Persuade other agent to a desirable state
 - Guarantee negotiation stage
 - Bind commitments at the agreed state



A general negotiation protocol

- ◆ 1) ranks states according to expected utility payoffs as an offering set.
- ◆ 2) Repeat the compensation negotiation until succeed or the set is empty.
 - 2.1) Agent i offers a particular state with/without proper compensation,
 - 2.2) Agent $-i$ accepts the offer or proposes a counter offer.
 - 2.3) The compensation negotiation succeeds with a compromise or fails when no more offers can be suggested by both sides .



Cont'd

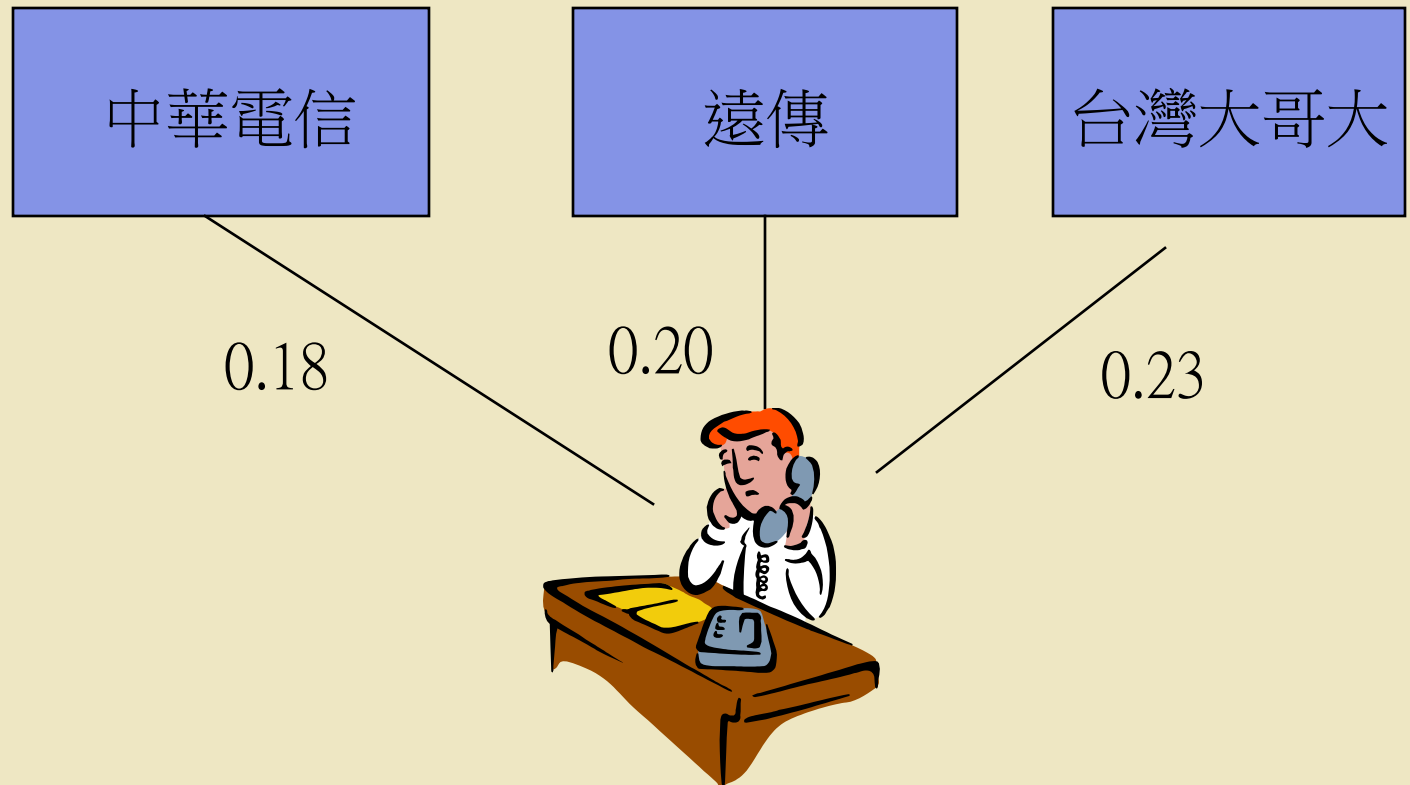
- ◆ 3) Repeat the guarantee negotiation
 - ◆ 3.1) Agent i makes guarantee offer for the state in order to bind the commitment from both sides if there is a profitable betraying action based on the expected monetary payoffs.
 - ◆ 3.2) Agent $-i$ accepts or proposes a counter offer.
 - ◆ 3.3) The guarantee negotiation ends with a compromise or a failure.



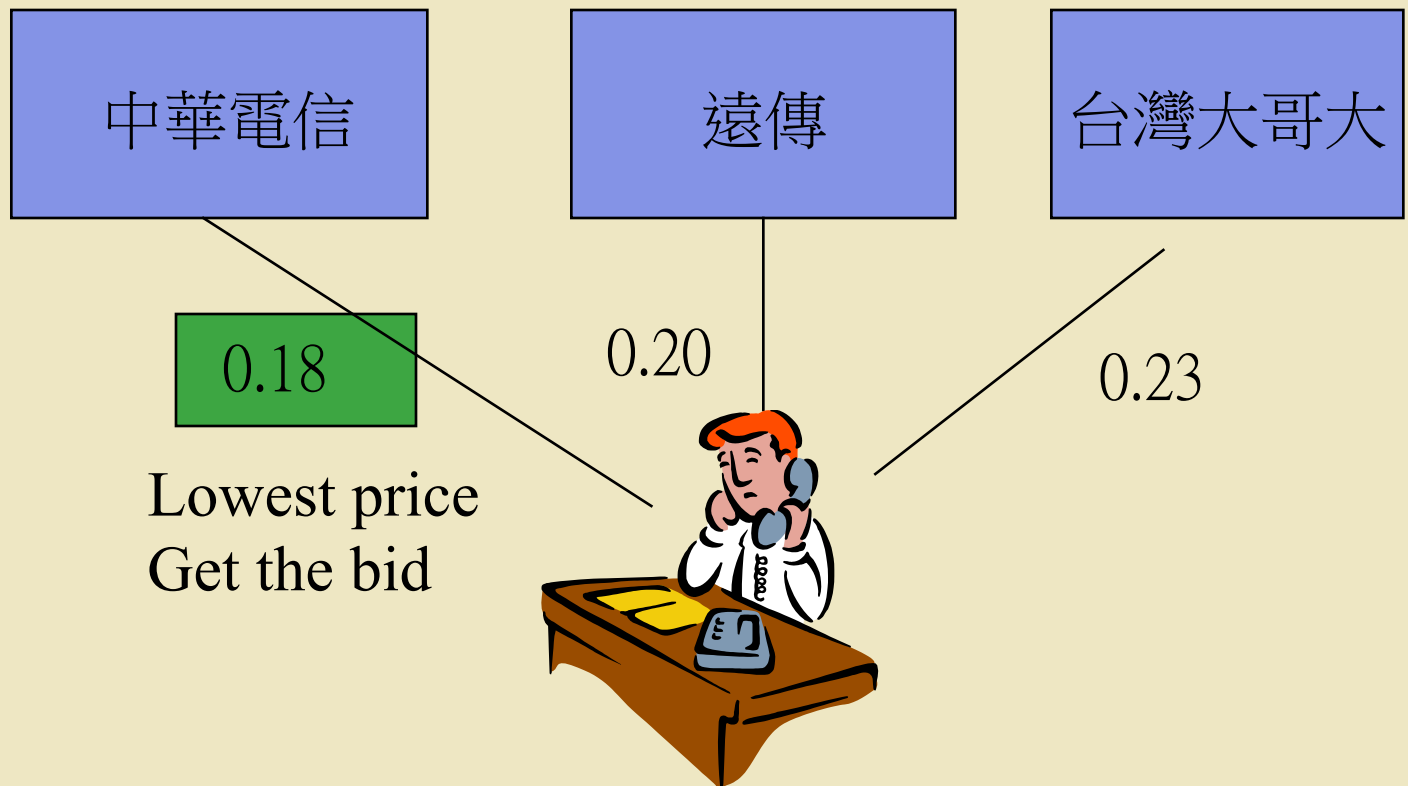
Outlines

- ◆ Introduction
- ◆ Contract Net Protocol
- ◆ Task oriented domain negotiation mechanisms
- ◆ Trusted third party mediated game theoretic negotiation
- Market oriented resource allocations

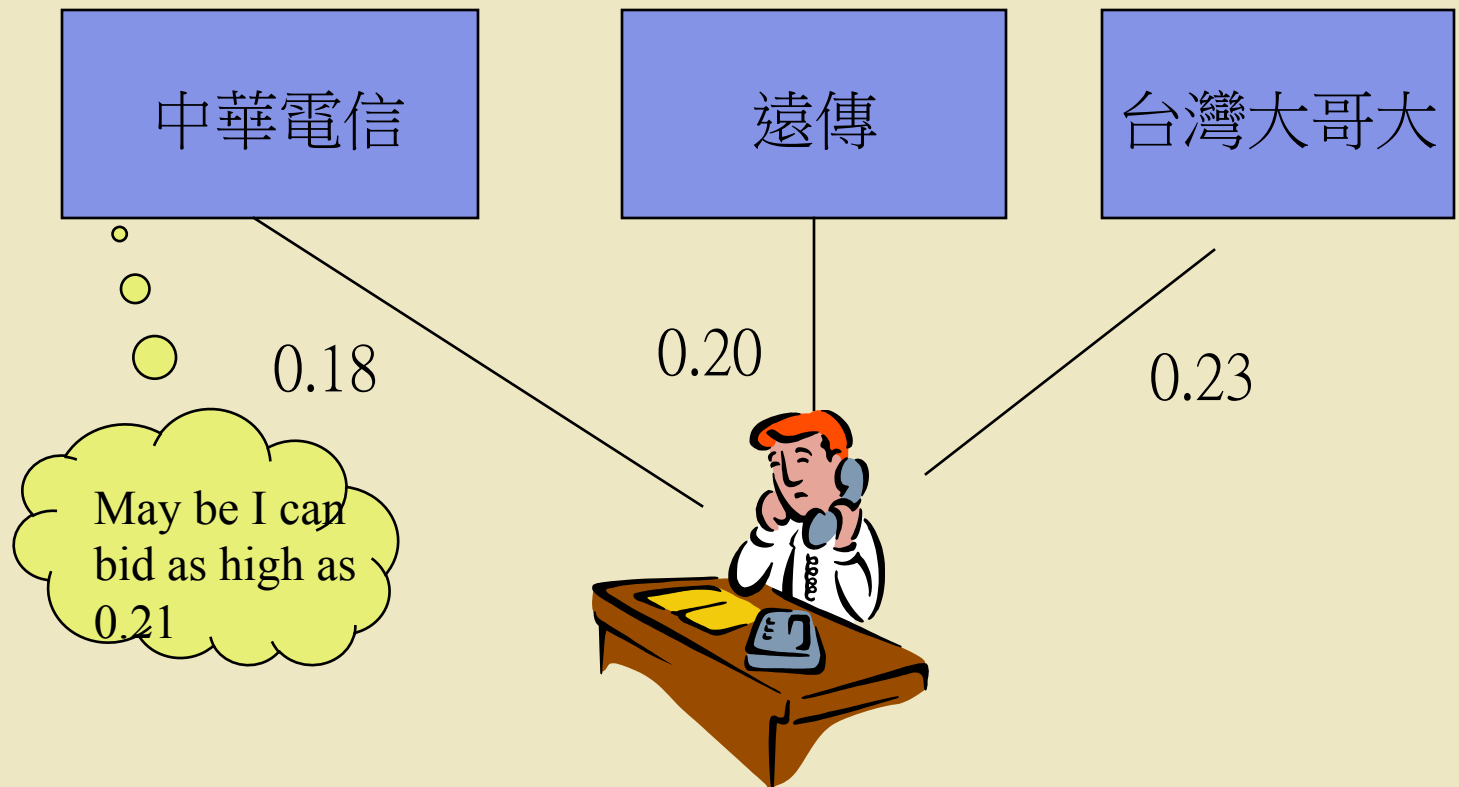
Telephone call competition example



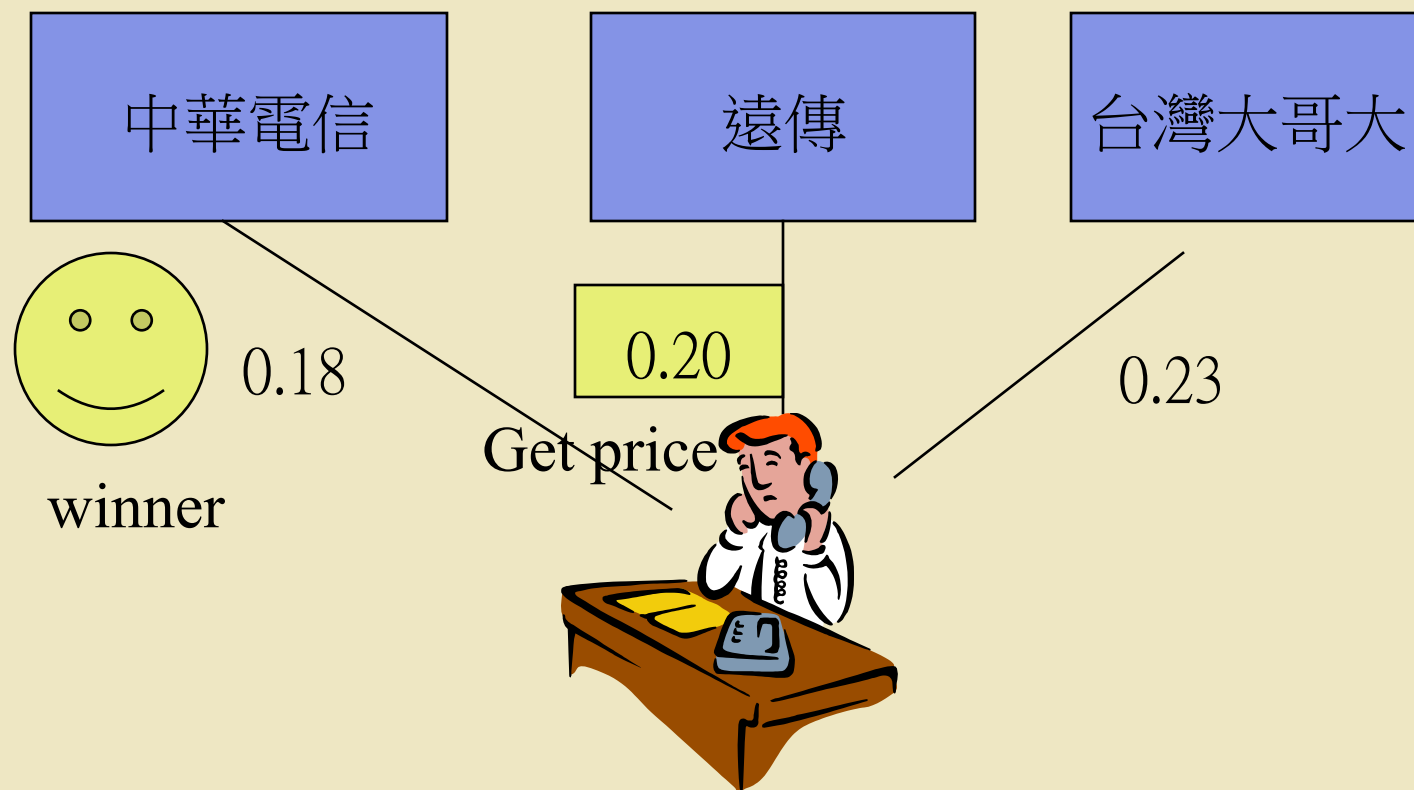
Telephone call competition



Strategic behaviors



Best bid win, gets second price





Vickrey's Auction mechanism

- ◆ No incentive for bidders to overbid or underbid: enforce a truthful bidding
- ◆ Over bid, then the company will have the risk to lose the bid that she could actually win.
- ◆ Under bid, then some other company might enter the gap, so eventually it will get the lower price than her acceptable price



General equilibrium based on market mechanism

- ◆ Consumers, producers, commodities
- ◆ Price mechanism
- ◆ Conditions for existence of a general equilibrium
- ◆ Conditions for uniqueness of a general equilibrium
- ◆ Algorithms to reach the Walrasian equilibrium



General (Walrasian) equilibrium

- ◆ An allocation $(x_1^*, \dots, x_I^*, y_1^*, \dots, Y_J^*)$ and a price vector $p = (p_1, \dots, p_L)$ constitute a Walrasian (general, competitive) equilibrium if
 - ◆ 1) for every j , y_j^* maximizes profits in Y_j , i.e.
$$p \bullet y_j \leq p \bullet y_j^*$$
 - ◆ 2) For every i , x_i^* is maximal for the preference of i in the budget set
$$\{x_i \in X_i : p \bullet x_i \leq p \bullet e_i + \sum_j \theta_{ij} p \bullet y_j^*\}$$
 - ◆ 3) Supply equals demand for every commodity I , i.e.
$$\sum_j x_{ij}^* = \sum_j e_{ij} + \sum_j y_{ij}^*$$



First Fundamental Theory of Welfare Economics

- ◆ Any Walrasian equilibrium allocation is Pareto optimal.



Second Fundamental Theorem of Welfare Economics

- ◆ Assume each consumer's preferences are convex, continuous, non-decreasing and locally insatiable and production sets are convex. Let (x^*, y^*) be an optimal allocation
- ◆ Then (x^*, y^*) is a Walarsian equilibrium



Applications of general equilibrium

- ◆ File storage allocation
- ◆ Network flow routing
- ◆ Distributed design
- ◆ Mirror site allocation
- ◆ Power load management
- ◆ ...



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